



HK IMO Training 2022-2023

Problem Set 9 Suggested Solutions



Problem 33 (Tuymaada Olympiad 2018 Junior Problem 1). Let a, b, c be given real numbers where $a \neq 0$. Prove that there is a polynomial $P(x)$ with real coefficients such that $x^2 + 1$ is a factor of $aP(x)^2 + bP(x) + c$.

Solution. Let $u + vi$ be a root of $at^2 + bt + c = 0$ where $u, v \in \mathbb{R}$ (possibly $v = 0$). We claim that $P(x) = u + vx$ satisfies the requirement. We check that

$$aP(i)^2 + bP(i) + c = a(u + vi)^2 + b(u + vi) + c = 0$$

and

$$aP(-i)^2 + bP(-i) + c = a(u - vi)^2 + b(u - vi) + c = \overline{a(u + vi)^2 + b(u + vi) + c} = 0.$$

Thus, both $x - i$ and $x + i$ are factors of $aP(x)^2 + bP(x) + c$, and it is divisible by $(x - i)(x + i) = x^2 + 1$. □

Problem 34 (JBMO 2016 Problem 3). Find all integers a, b, c, n such that $\frac{(a-b)(b-c)(c-a)}{2} + 2 = 2016^n$.

Solution. It is clear that $n \geq 0$. If $n \geq 1$, we have $(a-b)(b-c)(c-a) = 2(2016^n - 2) \equiv 3 \pmod{7}$. Let $x = a - b$, $y = b - c$ and $z = c - a$. Then $x + y + z = 0$ and $xyz \equiv 3 \pmod{7}$. This implies

$$x^3 + y^3 + z^3 = 3xyz + (x + y + z)(x^2 + y^2 + z^2 - xy - yz - zx) = 3xyz \equiv 2 \pmod{7}.$$

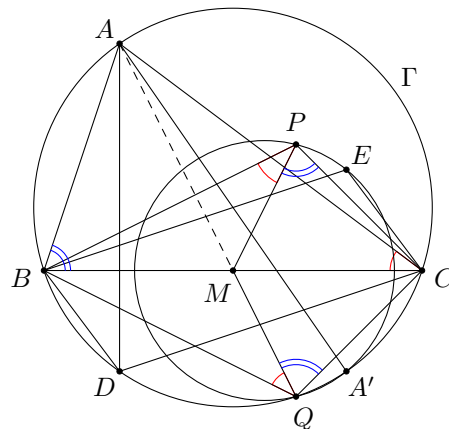
As $t^3 \equiv -1, 0, 1 \pmod{7}$, the numbers x^3, y^3, z^3 must be congruent to 0, 1, 1 modulo 7 in some order. WLOG assume $7 \mid x$. Then $y^3 \equiv z^3 \equiv 1 \pmod{7}$ and $7 \mid y + z$. But this is impossible since

$$1 \equiv z^3 \equiv (-y)^3 = -y^3 \equiv -1 \pmod{7}.$$

Therefore, there is no solution when $n \geq 1$.

Lastly, when $n = 0$, we need to solve $(a-b)(b-c)(c-a) = -2$, i.e. $xyz = -2$ with $x + y + z = 0$. WLOG assume $x \geq y \geq z$. As x, y, z are integers, the only solutions to $xyz = -2$ are $(2, 1, -1), (1, 1, -2), (-1, -1, -2)$. The only triple satisfying $x + y + z = 0$ is $(1, 1, -2)$. Therefore, we have $(a-b, b-c, c-a) = (1, 1, -2), (1, -2, 1), (-2, 1, 1)$. This means $(a, b, c) = (k+2, k+1, k), (k+1, k, k+2), (k, k+2, k+1)$ for some integer k . □

Problem 35 (JBMO 2017 Problem 3). In acute $\triangle ABC$, $AB \neq AC$. Let Γ be the circumcircle of $\triangle ABC$, and let AA' be a diameter of Γ . Let D be the point on Γ such that $AD \perp BC$. A point E is chosen such that $BDCE$ forms a parallelogram. Let M be the midpoint of BC . A point P is chosen on the same side of BC as A such that $\angle BPM = \angle ACB$ and $\angle CPM = \angle ABC$. Let Q be the second intersection point of Γ and the circumcircle of $\triangle A'EP$. Prove that A, M, Q are collinear.



Solution. Note that $BDA'C$ is an isosceles trapezoid since $\angle BAD = \angle CAA'$. Therefore, $A'CEB$ is a kite, where E and A' are symmetric about BC . Let Q' be the reflection of P in BC . Then $EPQ'A'$ is an isosceles trapezoid, so that it is cyclic. Also, we have

$$\angle BQ'C = \angle BQ'M + \angle CQ'M = \angle BPM + \angle CPM = \angle ACB + \angle ABC = 180^\circ - \angle BAC.$$

This implies Q' lies on Γ . Therefore, $Q' = Q$. Lastly, as

$$\angle MQB = \angle MPB = \angle ACB = \angle AQB,$$

we see that A, M, Q are collinear. □

Problem 36 (European Mathematical Cup 2015 Senior Problem 4). Let k be a positive integer and let G be a simple directed graph. Suppose each vertex has in-degree at least $3k + 1$ or has out-degree at least $3k + 1$ (or both). Prove that we can partition G into two nonempty subgraphs such that each vertex in each subgraph has in-degree at least k or has out-degree at least k .

Solution. For any subset X of the vertex set of G , we use $G(X)$ to denote the induced subgraph of G whose vertex set is X . We use $i_{G(X)}(v)$ and $o_{G(X)}(v)$ to denote the in-degree and the out-degree of a vertex v in $G(X)$. We say that X is *good* if $i_{G(X)}(v) \geq k$ or $o_{G(X)}(v) \geq k$ for each $v \in X$. Our goal is to partition G into two nonempty subgraphs X and Y such that both X and Y are good. We first prove a simple result.

Claim: If there exist disjoint nonempty subsets X and Y of the vertex set of G such that both X and Y are good, then we are done.

Proof. We choose such a pair (X, Y) so that the number of vertices in $X \cup Y$ is maximized. If $X \cup Y$ is the vertex set of G , we are done. Suppose on the contrary that $Z = G - (X \cup Y)$ is nonempty. Note that $Y \cup Z$ cannot be good due to the choice of (X, Y) . Therefore, there exists $v \in Z$ such that $i_{G(Y \cup Z)}(v) < k$ and $o_{G(Y \cup Z)}(v) < k$. Consider $W = X \cup \{v\}$. Then we have $i_{G(W)}(v) > (3k + 1) - k > k$ or $o_{G(W)}(v) > (3k + 1) - k > k$. It follows that both W and Y are good, contradicting the choice of (X, Y) . This proves the claim. □

Now, we say that a set of vertices X is *bad* if for every subset Y of X , we can find a vertex $v \in Y$ such that $i_{G(Y)}(v) \leq k$ and $o_{G(Y)}(v) \leq k$.

- If we cannot partition the vertex set of G into two nonempty subsets X and Y such that both X and Y are good, we consider a set of vertices A such that it is good and it has the minimum number of elements. We first show that A is bad. Firstly, every proper subset A' of A cannot be good. This means there exists $v \in A'$ such that $i_{G(A')}(v) < k$ and $o_{G(A')}(v) < k$. Also, for the set A itself, we pick any $w \in A$. Since $A - \{w\}$ is not good, there exists $x \in A - \{w\}$ such that $i_{G(A - \{w\})}(x) < k$ and $o_{G(A - \{w\})}(x) < k$. It follows that $i_{G(A)}(x) \leq k$ and $o_{G(A)}(x) \leq k$. By definition, A is bad.

Let B be the set of vertices which do not belong to A . By the assumption of this case, we know that B is not bad. Thus, there exists a subset B' of B such that $i_{G(B')}(y) > k$ or $o_{G(B')}(y) > k$ for all $y \in B'$. This shows B' is good. Now, as A and B' are disjoint and both of them are good, we are done by the claim.

- If we can partition the vertex set of G into two nonempty subsets A and B such that both A and B are bad, we choose such a partition which maximizes the sum of the number of edges in A and the number of edges in B . By the claim, we are done if each of A and B has a subset which is good. WLOG assume A has no good subset.

As B is bad, there exists $v \in B$ such that $i_{G(B)}(v) \leq k$ and $o_{G(B)}(v) \leq k$. Clearly, the set $B' = B - \{v\}$ is still bad. We claim that $A' = A \cup \{v\}$ is also bad. Indeed, consider any subset A'' of A' . If $v \notin A''$, it is clear that there exists $w \in A''$ such that $i_{G(A'')}(w) \leq k$ and $o_{G(A'')}(w) \leq k$ since $A'' \subset A$. If $v \in A''$, since the set $A'' - \{v\}$ is not good, there exists $x \in A'' - \{v\}$ such that $i_{G(A'' - \{v\})}(x) < k$ and $o_{G(A'' - \{v\})}(x) < k$. This implies $i_{G(A'')}(x) \leq k$ and $o_{G(A'')}(x) \leq k$. This proves A' is bad. Thus, the pair (A', B') is another partition of the vertex set into bad subsets. Note that B' is nonempty since there exists $y \in A$ such that $i_{G(A)}(y) \leq k$ and $o_{G(A)}(y) \leq k$, and hence B contains at least $(3k + 1) - k > 1$ vertices.

When we move v from B to A , at most $2k$ edges in B are removed since $i_{G(B)}(v) \leq k$ and $o_{G(B)}(v) \leq k$. But then at least $(3k + 1) - k = 2k + 1$ edges are formed in A' since v has in-degree or out-degree $\geq 3k + 1$ in G . This shows the total number of edges is increased, contradicting the choice of A and B . The proof is complete as we have handled all possible cases. □